

YIZHOU DING

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EDUCATION

Nanyang Technological University

Singapore

MSc in Financial Engineering

2026 – Present

- **Graduate coursework (completed / scheduled):** Stochastic Calculus in Finance, Derivative Securities, Financial Time Series Analysis, Optimization in Finance, Numerical Methods for Financial Instrument Pricing, Artificial Intelligence Techniques in Finance, Object-Oriented Programming I–II.

Southwestern University of Finance and Economics (SWUFE) & University of Delaware (UD)

Chengdu, China

Dual Bachelor's Degrees in Financial Mathematics & Finance; GPA: 90.75/100

2022 – 2026

- **Undergraduate coursework:** Stochastic Processes, Financial Stochastic Analysis, Probability Theory, Mathematical Statistics, Ordinary Differential Equations, Econometrics, Machine Learning.

RESEARCH EXPERIENCE

A Variable-Step Crank–Nicolson and High-Order Compact Scheme for a Nonlinear Black–Scholes Type Equation

Chengdu, China

Research Assistant, Supervisor: Prof. Pin Lyu, SWUFE; **First author**; manuscript under review

Sep. 2024 – Present

- Worked on a fully discrete scheme combining variable-step Crank–Nicolson time approximation, fourth-order compact spatial discretization, and explicit treatment of nonlinear terms for a Black–Scholes type equation with counterparty risk.
- Worked through key components of the stability and convergence analysis using discrete energy estimates, contributing to the optimal error bound $\mathcal{O}(\tau^2 + h^4)$ and comparing alternative proof arguments for mathematical validity.
- Implemented MATLAB experiments to verify convergence rates, stability, and computational efficiency across mesh and time-step settings; led preparation of the resulting manuscript as first author.

Automatic Event Scheduling Based on LLMs

Chengdu, China

Team Project Leader, Instructor: Prof. Leting Zhang, UD

Feb. 2025 – Apr. 2025

- Built an end-to-end scheduling prototype combining an SQL venue database with LLM interpretation of natural-language constraints; localized DeepSeek models through Ollama and developed a Flask interface.
- Selected by the instructor as an exemplary project for future cohorts, demonstrating implementation from data organization and model integration to a usable application.

People's Bank of China: Global Central Bank Digital Currency Study

Chengdu, China

Research Contributor

Jun. 2023 – Dec. 2023

- Synthesized global CBDC initiatives and the Digital Euro investigation phase, connecting technology design and adoption mechanisms with policy, regulation, payment infrastructure, and financial intermediation.

SELECTED QUANTITATIVE EXPERIENCE

SWS Futures

Shanghai, China

Quantitative Research Intern

Jul. 2025 – Sep. 2025

- Extended the Nelson–Siegel yield-curve framework with multidimensional macroeconomic factors and XGBoost to test whether macro information explained movements beyond conventional latent factors.
- Refined a moving-average RSI signal through standardization, hyperparameter tuning, and systematic backtesting diagnostics, evaluating robustness and transferability across agricultural futures contracts.

Galaxy Futures

Shanghai, China

Futures Analyst Intern

Dec. 2024 – May 2025

- Formulated and tested hypotheses on intraday volatility patterns using high-frequency crude-oil and soybean-meal options data, separating empirical pattern identification from strategy evaluation under explicit backtesting assumptions.
- Developed a localized LLM-assisted forecasting workflow integrating price-volume indicators, sentiment, fundamentals, and macroeconomic variables into a coherent modelling pipeline.
- Compared minimum-variance futures hedging, one-to-one futures hedging, and options-based hedging within a consistent framework, evaluating trade-offs in hedging effectiveness, cost, and implementation for rebar exposure.

Futurus Vessel Capital

Shanghai, China

Risk Analyst Intern

Jul. 2024 – Aug. 2024

- Implemented reusable Python routines for VaR, drawdown, annualized return, and scenario-based stress testing, and applied ARIMA, volatility proxies, and simulation methods to financial time series across market segments.

ADDITIONAL ACADEMIC EXPERIENCE

Database Design & Implementation

Chengdu, China

Teaching Assistant, Supervisor: Prof. Leting Zhang, UD

Feb. 2026 – Jun. 2026

- Supported instruction in relational database design and SQL, troubleshooting implementation issues in assignments and projects and assisting with grading and feedback.

TECHNICAL SKILLS & SELECTED HONORS

Programming: Python, MATLAB, R, C++, SQL **Research/Tools:** LaTeX, Git, EViews, Flask, Ollama, Codex, Zotero

Languages: Mandarin Chinese (native); English (IELTS 7.5; GRE 326)

Honors: “Xuechuang” Cup, National Second Prize (2024); Academic Scholarship, Top 10% (2022–2026)

ACADEMIC REFERENCE

Prof. Fei Xie, University of Delaware | xief@udel.edu | Faculty Website